

Quick Check

Lesson 2: Stepwise Growth: Why Hard-Shelled Animals Grow in Jumps

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.2: Growth and Development in Animals | Lesson 2

A grasshopper nymph in a school garden in Kisumu is hatched and its body length is measured every few days for 10 weeks. Its body is covered by a hard exoskeleton. What shape should the graph of length against time have?

- A** A smooth curve that rises steadily, like a child's height
- B** A staircase shape with flat parts (plateaus) and sudden rises
- C** A flat line, because insects do not grow after hatching
- D** A straight line sloping upwards from the first day

A locust nymph's length stayed at 14 mm from Day 12 to Day 17. On Day 18 it measured 19 mm. By how many millimetres did its length increase between Day 17 and Day 18?

A 5 mm

B 19 mm

C 14 mm

D 33 mm

Learners plot both length and mass of a desert locust nymph over 35 days. Which statement best describes what they are likely to see between moults?

- A** Length and mass both stay flat the whole time until the adult stage
- B** Length rises smoothly while mass stays flat, then mass jumps at a moult
- C** Mass can keep increasing gradually while length stays flat, then length jumps at a moult
- D** Both length and mass drop each time the nymph feeds

A grasshopper nymph keeps eating sukuma wiki, but its length does not change from Day 12 to Day 17. What is the best scientific reason?

- A** It stops digesting food during those days
- B** Its hard exoskeleton cannot stretch, so its length cannot increase until it moults
- C** The nymph shrinks slightly and then regrows
- D** The old skin is thin and peels off slowly like sunburned skin

A learner finds an empty exuvia on a maize leaf and says, "It is just a thin peel of skin, like sunburned skin." Which statement corrects this idea?

- A** The exuvia is the new skin which has not hardened yet
- B** The exuvia is a dead young insect that did not survive
- C** The exuvia is a thin peel that comes off only from the surface of the body
- D** The exuvia is a complete hard outer casing, shaped like the insect with legs, left behind when the insect moults

A newly emerged grasshopper is larger than the exuvia it has just left behind. Which explanation is correct?

- A** The new cuticle hardened first and then the insect grew inside it
- B** The old exuvia shrank after it was shed
- C** The old exoskeleton stretched slowly as the grasshopper grew inside it
- D** After splitting the old cuticle, the insect swells and expands while the new cuticle is still soft, before it hardens

A grasshopper nymph is somehow unable to moult. Based on the lesson, what would happen to its body length?

- A** It would stay limited by the hard exoskeleton and could not increase
- B** It would increase in steps without needing a new cuticle
- C** It would grow only in mass and shrink in length
- D** It would grow smoothly because the exoskeleton would stretch

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).